

BELO HORIZONTE PAMPULHA, BrazilWMO: **835830**Lat: **19.851S**Lon: **43.951W**Elev: **2589**StdP: **13.37**Time Zone: **-3.00 (SAB)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.8	53.7	38.8	38.2	77.6	41.3	42.0	75.2	15.9	70.5	14.3	70.5	1.9	100	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.8	91.5	67.4	89.7	67.6	88.0	67.8	72.7	82.9	71.9	82.2	71.3	81.2	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	122.2	74.6	69.7	120.4	74.3	68.4	114.8	73.9	38.2	83.5	37.5	81.8	36.8	81.8	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.8	11.7	DB	47.3	95.3	2.1	1.6	45.8	96.5	44.6	97.4	43.4	98.3	41.8	99.5	(4)
(5)			WB	44.5	75.2	2.4	2.6	42.8	77.1	41.4	78.7	40.0	80.1	38.3	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	76.8	77.5	75.8	73.3	69.4	67.2	66.9	69.1	72.8	74.9	74.4	75.5	(6)	
	DBStd	4.82	2.98	2.60	2.78	3.04	3.00	2.87	3.05	3.28	4.13	4.14	3.86	2.84	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	39	0	0	0	0	3	13	16	5	1	1	0	0	(9)	
	CDD50	8313	830	769	800	698	601	516	525	594	684	773	733	790	(10)	
	CDD65	2881	365	349	335	248	139	79	77	134	236	310	284	325	(11)	
	CDH74	20188	2751	2742	2221	1487	780	542	608	1077	1875	2297	1784	2024	(12)	
	CDH80	6130	980	1003	686	322	102	51	68	255	659	828	562	614	(13)	
(14) Wind	WSAvg	5.2	5.0	5.3	4.8	4.7	4.5	4.2	4.7	5.7	6.4	6.4	5.5	4.8	(14)	
(15) Precipitation	PrecAvg	57.2	11.6	6.1	6.7	2.5	1.1	0.4	0.2	0.4	1.9	4.0	9.1	13.2	(15)	
	PrecMax	79.3	27.4	13.7	13.2	7.5	3.5	2.4	1.5	2.2	4.6	13.4	15.2	27.4	(16)	
	PrecMin	32.6	3.3	0.7	1.9	0.5	0.0	0.0	0.0	0.0	0.0	1.1	4.2	5.2	(17)	
	PrecStd	10.6	6.1	3.3	3.1	1.6	0.9	0.6	0.4	0.6	1.3	2.6	3.0	5.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	92.9	91.1	88.1	84.6	83.9	84.3	88.2	93.4	93.6	92.8	89.9	(19)	
		MCWB	68.7	71.0	70.2	69.7	66.6	63.7	62.4	62.0	64.2	65.1	68.3	68.5	(20)	
	2%	DB	89.7	89.9	88.0	85.8	82.6	81.0	82.1	85.7	89.8	91.0	88.2	87.9	(21)	
		MCWB	69.1	69.8	70.0	68.9	65.5	63.2	61.5	61.8	63.4	65.4	67.9	69.1	(22)	
	5%	DB	87.8	88.0	86.1	83.9	80.8	79.2	80.2	82.7	86.6	87.8	86.0	86.1	(23)	
		MCWB	68.8	69.3	69.6	68.4	65.0	62.8	61.4	61.1	63.5	65.7	67.9	68.7	(24)	
	10%	DB	85.8	86.1	84.2	82.0	78.8	77.2	77.4	80.4	84.0	84.6	82.8	84.0	(25)	
		MCWB	68.7	68.7	69.5	67.8	64.8	62.2	61.2	61.1	63.7	66.0	67.7	68.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.6	73.7	73.4	72.2	69.9	67.8	65.7	66.4	70.0	71.1	71.9	73.1	(27)	
		MCDB	82.9	85.3	83.6	81.7	79.3	78.7	77.8	79.5	82.4	81.2	82.3	82.3	(28)	
	2%	WB	72.2	72.4	72.0	71.0	68.3	65.8	64.4	65.0	68.2	69.8	70.9	71.9	(29)	
		MCDB	82.2	84.0	82.6	80.8	77.8	76.2	76.9	78.3	80.0	80.4	80.6	80.7	(30)	
	5%	WB	71.4	71.5	71.3	70.0	67.3	64.7	63.4	63.7	66.7	68.9	70.0	71.1	(31)	
		MCDB	81.3	82.8	81.6	79.3	76.6	74.8	75.5	76.8	78.3	79.5	79.0	79.5	(32)	
	10%	WB	70.5	70.7	70.5	69.2	66.1	63.5	62.4	62.6	65.7	68.1	69.2	70.4	(33)	
		MCDB	79.8	80.9	80.0	78.1	74.9	73.5	74.2	75.4	77.3	78.7	78.0	78.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	17.8	16.4	17.8	19.1	21.3	22.2	22.5	20.7	18.4	15.7	15.7	(35)	
		MCDBR	20.4	20.6	19.7	19.8	21.7	24.0	25.7	26.6	25.3	22.9	20.4	19.6	(36)	
	5% WB	MCWBR	6.0	5.8	6.1	6.8	7.9	8.9	9.0	8.2	7.0	6.7	6.2	6.2	(37)	
		MCWBR	16.8	17.9	16.8	17.4	18.5	20.9	21.5	21.8	19.0	18.2	16.1	15.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.383	0.378	0.357	0.338	0.326	0.330	0.362	0.446	0.440	0.403	0.396	(40)	
		taud	2.449	2.461	2.497	2.516	2.539	2.571	2.534	2.424	2.183	2.254	2.395	2.436	(41)	
	Ebn at Noon		302	301	295	289	283	281	283	283	270	281	297	301	(42)	
		Edh at Noon	39	38	35	32	29	27	29	34	47	46	41	39	(43)	
(44) All-Sky Solar Radiation	RadAvg		1850	1896	1672	1543	1339	1262	1348	1580	1683	1772	1633	1731	(44)	
	RadStd		272	207	150	99	76	82	76	105	173	207	154	181	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	-1.51	-1.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page